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Codetantra practical 3

3.1.1 problem statement

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using `ndim`
- The shape of the array using `shape`
- The total number of elements in the array using `size`

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, `rows` and `columns`, representing the number of rows and the number of columns.
- The next `rows` lines contain `columns` space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use `reshape()` function to reshape the input array with the specified number of rows and columns.

Code:

```
import numpy as np
r,c = map(int, input().split())
elements = []
for _ in range(r) :
    elements.extend(map(int, input().split()))
array = np.array(elements).reshape(r,c)

print(array)
```

```
print(array.ndim)
```

```
print(array.shape)
```

```
print(array.size)
```

Output:

Average time
0.010 s
9.83 ms



Maximum time
0.017 s
17.00 ms



✓ 2 out of 2 shown test case(s) passed

✓ 10 out of 10 hidden test case(s) passed

✓ Test case 1
17 ms



Expected output	Actual output
3 4	3 4
1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
9 10 11 12	9 10 11 12

<code>[[·1·2·3·4]</code>	<code>[[·1·2·3·4]</code>
<code>·[·5·6·7·8]</code>	<code>·[·5·6·7·8]</code>
<code>·[·9·10·11·12]]</code>	<code>·[·9·10·11·12]]</code>
<code>2</code>	<code>2</code>
<code>(3,·4)</code>	<code>(3,·4)</code>
<code>12</code>	<code>12</code>

✓ Test case 2 3 ms    	
Expected output	Actual output
<code>0 0</code>	<code>0 0</code>
<code>[]</code>	<code>[]</code>
<code>2</code>	<code>2</code>
<code>(0,·0)</code>	<code>(0,·0)</code>
<code>0</code>	<code>0</code>

Practical Lab Assignment

3.2.1 problem statement

The given code takes two 3×3 matrices, `matrix_a`, and `matrix_b`, as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. **Addition** (`matrix_a + matrix_b`)

2. **Subtraction** ($\text{matrix_a} - \text{matrix_b}$)

3. **Element-wise Multiplication** ($\text{matrix_a} * \text{matrix_b}$)

4. **Matrix Multiplication** ($\text{matrix_a} . \text{matrix_b}$)

5. **Transpose of Matrix A**

Input Format:

- The user will input 3 rows for , each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for , each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.
5. The Transpose of Matrix A.

Code:

```
import numpy as np
```

```
# Input matrices
```

```
print("Enter Matrix A:")
```

```
matrix_a = np.array([list(map(int, input().split())) for i  
in range(3)])
```

```
print("Enter Matrix B:")
```

```
matrix_b = np.array([list(map(int, input().split())) for i  
in range(3)])
```

```
# Addition
```

```
addition_result = matrix_a + matrix_b
```

```
print("Addition (A + B):")
```

```
print(addition_result)
```

```
# Subtraction
```

```
subtraction_result = matrix_a - matrix_b
```

```
print("Subtraction (A - B):")
```

```
print(subtraction_result)
```

```
# Multiplication (element-wise)
```

```
elementwise_multiplication_result = matrix_a *  
matrix_b
```

```
print("Element-wise Multiplication (A * B):")
print(elementwise_multiplication_result)
# Matrix multiplication (dot product)
matrix_multiplication_result = np.dot(matrix_a,
matrix_b)
print("A dot B:")
print(matrix_multiplication_result)
# Transpose
transpose_a = matrix_a.T
print("Transpose of A:")
print(transpose_a)
```

Average time	Maximum time
0.019 s 19.50 ms	0.027 s 27.00 ms

✓ 2 out of 2 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1
27 ms

Expected output Actual output

Enter Matrix A: ↵	Enter Matrix A: ↵
1 2 3	1 2 3
4 5 6	4 5 6
7 8 9	7 8 9
Enter Matrix B: ↵	Enter Matrix B: ↵
1 1 1	1 1 1
1 1 1	1 1 1
1 1 1	1 1 1
Addition (A + B): ↵	Addition (A + B): ↵
[[2 3 4] ↵	[[2 3 4] ↵
· [5 6 7] ↵	· [5 6 7] ↵
· [8 9 10]] ↵	· [8 9 10]] ↵
Subtraction (A - B): ↵	Subtraction (A - B): ↵
[[0 1 2] ↵	[[0 1 2] ↵
· [3 4 5] ↵	· [3 4 5] ↵
· [6 7 8]] ↵	· [6 7 8]] ↵
Element-wise Multiplication (A * B): ↵	Element-wise Multi plication (A * B): ↵
[[1 2 3] ↵	[[1 2 3] ↵
· [4 5 6] ↵	· [4 5 6] ↵
· [7 8 9]] ↵	· [7 8 9]] ↵
A dot B: ↵	A dot B: ↵
[[6 6 6] ↵	[[6 6 6] ↵
· [15 15 15] ↵	· [15 15 15] ↵

· [24·24·24]] ↗

· [24·24·24]] ↗

Transpose of A: ↗

Transpose of A: ↗

[[1·4·7] ↗

[[1·4·7] ↗

· [2·5·8] ↗

· [2·5·8] ↗

· [3·6·9]] ↗

· [3·6·9]] ↗

✓ Test case 2

18 ms



Expected output

Actual output

Enter Matrix A: ↗

Enter Matrix A: ↗

2 4 8

2 4 8

9 6 5

9 6 5

7 8 9

7 8 9

Enter Matrix B: ↗

Enter Matrix B: ↗

10 12 13

10 12 13

14 15 16

14 15 16

17 18 19

17 18 19

Addition (A+B): ↗

Addition (A+B):
↗

[[12·16·21] ↗

[[12·16·21] ↗

· [23·21·21] ↗

· [23·21·21] ↗

· [24·26·28]] ↗

· [24·26·28]] ↗

Subtraction (A-B):
↗

Subtraction (A-B):
↗

[[·-8·-8·-5] ↗

[[·-8·-8·-5] ↗

· [·-5·-9·-11] ↗

· [·-5·-9·-11] ↗

<code>· [-10 · -10 · -10]]</code>	<code>· [-10 · -10 · -10]]</code>
Element-wise Multiplication (A.*B):	Element-wise Multiplication (A.*B):
<code>[[· 20 · 48 · 104]</code>	<code>[[· 20 · 48 · 104]</code>
<code>· [126 · 90 · 80]</code>	<code>· [126 · 90 · 80]</code>
<code>· [119 · 144 · 171]]</code>	<code>· [119 · 144 · 171]]</code>
A.dot.B:	A.dot.B:
<code>[[212 · 228 · 242]</code>	<code>[[212 · 228 · 242]</code>
<code>· [259 · 288 · 308]</code>	<code>· [259 · 288 · 308]</code>
<code>· [335 · 366 · 390]]</code>	<code>· [335 · 366 · 390]]</code>
Transpose of A:	Transpose of A:
<code>[[2 · 9 · 7]</code>	<code>[[2 · 9 · 7]</code>
<code>· [4 · 6 · 8]</code>	<code>· [4 · 6 · 8]</code>
<code>· [8 · 5 · 9]]</code>	<code>· [8 · 5 · 9]]</code>

3.2.2 problem statement

You are given two arrays arr1 and arr2. You need to perform horizontal and vertical stacking operations on them using NumPy.

- **Horizontal Stacking:** Stack the two matrices horizontally (side by side).
- **Vertical Stacking:** Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Code:

```
import numpy as np
```

```
# Input matrices
```

```
print("Enter Array1:")
```

```
arr1 = np.array([list(map(int, input().split())) for i in  
range(3)])
```

```
print("Enter Array2:")
```

```
arr2 = np.array([list(map(int, input().split())) for i in  
range(3)])
```

```
# Perform horizontal stacking (hstack)
```

```
a=np.hstack((arr1,arr2))
```

```
print("Horizontal Stack:")
```

```
print(a)
```

```
# Perform vertical stacking (vstack)
```

```
b=np.vstack((arr1,arr2))
```

```
print("Vertical Stack:")
```

```
print(b)
```

Output:

Average time 0.013 s 12.50 ms	Maximum time 0.017 s 17.00 ms
✓ 2 out of 2 shown test case(s) passed	
✓ 2 out of 2 hidden test case(s) passed	
Test case 1 ✓ 17 ms	
Expected output	Actual output

Enter Array1: ↵	Enter Array1: ↵
1 2 3	1 2 3
4 5 6	4 5 6
7 8 9	7 8 9
Enter Array2: ↵	Enter Array2: ↵
4 5 6	4 5 6
7 8 9	7 8 9
10 11 12	10 11 12
Horizontal Stack: ↵	Horizontal Stack: ↵
[[1 2 3 4 5 6] ↵	[[1 2 3 4 5 6] ↵
[4 5 6 7 8 9] ↵	[4 5 6 7 8 9] ↵
[7 8 9 10 11 12]] ↵	[7 8 9 10 11 12]] ↵
Vertical Stack: ↵	Vertical Stack: ↵
[[1 2 3] ↵	[[1 2 3] ↵
[4 5 6] ↵	[4 5 6] ↵
[7 8 9] ↵	[7 8 9] ↵
[4 5 6] ↵	[4 5 6] ↵
[7 8 9] ↵	[7 8 9] ↵
[10 11 12]] ↵	[10 11 12]] ↵

Test case 2
12 ms

Expected output
Actual output

Enter Array1: ↵	Enter Array1: ↵
-1 -2 -3	-1 -2 -3
-4 -5 -6	-4 -5 -6
-7 -8 -9	-7 -8 -9
Enter Array2: ↵	Enter Array2: ↵
10 11 12	10 11 12
13 14 15	13 14 15
16 17 18	16 17 18
Horizontal Stack: ↵	Horizontal Stack: ↵
[[-1 -2 -3 10 11 1 2] ↵	[[-1 -2 -3 10 11 1 2] ↵
· [-4 -5 -6 13 14 1 5] ↵	· [-4 -5 -6 13 14 1 5] ↵
· [-7 -8 -9 16 17 1 8]] ↵	· [-7 -8 -9 16 17 1 8]] ↵
Vertical Stack: ↵	Vertical Stack: ↵
[[-1 -2 -3] ↵	[[-1 -2 -3] ↵
· [-4 -5 -6] ↵	· [-4 -5 -6] ↵
· [-7 -8 -9] ↵	· [-7 -8 -9] ↵
· [10 11 12] ↵	· [10 11 12] ↵
· [13 14 15] ↵	· [13 14 15] ↵
· [16 17 18]] ↵	· [16 17 18]] ↵

3.2.3 problem statement

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.

- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

Code:

```
import numpy as np
```

```
# Take user input for the start, stop, and step of the sequence
```

```
start = int(input())
```

```
stop = int(input())
```

```
step = int(input())
```

```
# Generate the sequence using np.arange()
```

```
a=np.arange(start,stop,step)
```

```
print(a)
```

```
# Print the generated sequence
```

Output:

Average time

0.008 s

8.00 ms



Maximum time

0.013 s

13.00 ms



✓ 2 out of 2 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1

13 ms



Expected output	Actual output
3	3
10	10
2	2
[3·5·7·9]	[3·5·7·9]

Test case 2	
5 ms	
Expected output	Actual output
10	10
20	20
30	30
[10]	[10]

3.2.4 problem statement

You are given two arrays A and B. Your task is to complete the function `array_operations`, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:

- Compute the element-wise sum, difference, and product of the two arrays.

2. Statistical Operations:

- Calculate the mean, median, and standard deviation of array A.

3. Bitwise Operations:

- Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: $A_i \text{ OR } B_i$).

Input Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

Code:

```
import numpy as np
```

```
def array_operations(A, B):
```

```
    # Convert A and B to NumPy arrays
```

```
    A = np.array(A)
```

```
    B = np.array(B)
```

```
    # Arithmetic Operations
```

```
    sum_result = A + B
```

```
diff_result = A - B
```

```
prod_result = A * B
```

```
# Statistical Operations
```

```
mean_A = np.mean(A)
```

```
median_A = np.median(A)
```

```
std_dev_A = np.std(A)
```

```
# Bitwise Operations
```

```
and_result = A & B
```

```
or_result = A | B
```

```
xor_result = A ^ B
```

```
# Output results with one space between each  
element
```

```
print("Element-wise Sum:", ' '.join(map(str,  
sum_result)))
```

```
print("Element-wise Difference:", ' '.join(map(str,  
diff_result)))
```

```
print("Element-wise Product:", ' '.join(map(str,  
prod_result)))
```

```

print(f"Mean of A: {mean_A}")
print(f"Median of A: {median_A}")
print(f"Standard Deviation of A: {std_dev_A}")
    print("Bitwise AND:", ' '.join(map(str,
and_result)))
    print("Bitwise OR:", ' '.join(map(str, or_result)))
    print("Bitwise XOR:", ' '.join(map(str, xor_result)))

```

```

A = list(map(int, input().split())) # Elements of array A
B = list(map(int, input().split())) # Elements of array B
array_operations(A, B)

```

Output:

Average time	Maximum time
0.008 s 7.67 ms	0.011 s 11.00 ms




 1 out of 1 shown test case(s) passed

 2 out of 2 hidden test case(s) passed

 Test case 1
11 ms






Expected output
Actual output

1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
Element-wise Sum: 6 8 10 12	Element-wise Sum: 6 8 10 12
Element-wise Difference: -4 -4 -4 -4	Element-wise Difference: -4 -4 -4 -4
Element-wise Product: 5 12 21 32	Element-wise Product: 5 12 21 32
Mean of A: 2.5	Mean of A: 2.5
Median of A: 2.5	Median of A: 2.5
Standard Deviation of A: 1.118033988749895	Standard Deviation of A: 1.118033988749895
Bitwise AND: 1 2 3 0	Bitwise AND: 1 2 3 0
Bitwise OR: 5 6 7 12	Bitwise OR: 5 6 7 12
Bitwise XOR: 4 4 4 12	Bitwise XOR: 4 4 4 12

3.2.5 problem statement

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the original_array and assigning it to view_array.
- Creating a copy of the original_array and assigning it to copy_array.

After completing these steps, observe how modifying the view affects the `original_array`, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

Original array after modifying view: **<original_array>**

View array: **<view_array>**

- After modifying the copy:

Original array after modifying copy: **<original_array>**

Copy array: **<copy_array>**

Code:

```
import numpy as np
```

```
inputlist = list(map(int,input().split(" ")))
```

```
# Original array
original_array = np.array(inputlist)

# Create a view
view_array = original_array.view()

# Create a copy
copy_array = original_array.copy()

# Modify the view
view_array[0] = 99

print("Original array after modifying view:",
      original_array)

print("View array:", view_array)

# Modify the copy
copy_array[1] = 88

print("Original array after modifying copy:",
      original_array)

print("Copy array:", copy_array)
```

Output:

Average time

0.005 s

4.75 ms



Maximum time

0.010 s

10.00 ms



✓ 2 out of 2 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1

10 ms



Expected output

10 20 30 40 50 60
70 80

Actual output

10 20 30 40 50 60
70 80

Original array after
modifying view: [99
20 30 40 50 60 70 8
0]

Original array af
ter modifying vie
w: [99 20 30 40 5
0 60 70 80]

View array: [99 20 3
0 40 50 60 70 80]

View array: [99 2
0 30 40 50 60 70
80]

Original array after
modifying copy: [99
20 30 40 50 60 70 8
0]

Original array af
ter modifying cop
y: [99 20 30 40 5
0 60 70 80]

Copy array: [10 88 3
0 40 50 60 70 80]

Copy array: [10 8
8 30 40 50 60 70
80]

✓ Test case 2

3 ms



Expected output	Actual output
99 2 3 4 5	99 2 3 4 5
Original array after modifying view: [99 2 3 4 5]	Original array af ter modifying vie w: [99 2 3 4 5]
View array: [99 2 3 4 5]	View array: [99 2 3 4 5]
Original array after modifying copy: [99 2 3 4 5]	Original array af ter modifying cop y: [99 2 3 4 5]
Copy array: [99 88 3 4 5]	Copy array: [99 8 8 3 4 5]

3.2.6 problem statement

The given code in the editor takes a single array, array1, as space-separated integers as input from the user.

Additionally, it takes the following inputs:

- search_value: The value to search for in the array.
- count_value: The value to count its occurrences in the array.
- broadcast_value: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

- 1. Searching:** Find the indices where `search_value` appears in `array1` and print these indices.
- 2. Counting:** Count how many times `count_value` appears in `array1` and print the count.
- 3. Broadcasting:** Add `broadcast_value` to each element of `array1` using broadcasting, and print the resulting array.
- 4. Sorting:** Sort `array1` in ascending order and print the sorted array.

Input Format:

1. A single line containing space-separated integers representing `array1`.
2. An integer `search_value` represents the value to search for in the array.
3. An integer `count_value` represents the value to count in the array.
4. An integer `broadcast_value` represents the value to add to each element of the array.

Output Format:

1. The indices where `search_value` occurs in `array1`.

2. The count of occurrences of count_value in array1.
3. The array after adding the broadcast_value to each element.
4. The sorted array.

Code:

```
import numpy as np

# Input array from the user
array1 = np.array(list(map(int, input().split()))))

# Searching
search_value = int(input("Value to search: "))
count_value = int(input("Value to count: "))
broadcast_value = int(input("Value to add: "))

# Find indices where value matches in array1
a=np.where(array1==search_value)[0]
print(a)

# Count occurrences in array1
b=np.count_nonzero(array1==count_value)
```

```
print(b)

# Broadcasting addition

c=array1 + broadcast_value

print(c)

# Sort the first array

d=np.sort(array1)

print(d)
```

Output:

Average time

0.009 s

9.25 ms



Maximum time

0.013 s

13.00 ms



✓ 2 out of 2 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1

13 ms



Expected output	Actual output
1 1 1 2 2 2	1 1 1 2 2 2
Value to search: 1	Value to search: 1

Value to count: 2	Value to count: 2
Value to add: 2	Value to add: 2
[0 1 2]	[0 1 2]
3	3
[3 3 3 4 4 4]	[3 3 3 4 4 4]
[1 1 1 2 2 2]	[1 1 1 2 2 2]

✓ Test case 2 7 ms    	
Expected output	Actual output
1 2 3 4 5	1 2 3 4 5
Value to search: 6	Value to search: 6
Value to count: 6	Value to count: 6
Value to add: 6	Value to add: 6
[]	[]
0	0
[7 8 9 10 11]	[7 8 9 10 11]
[1 2 3 4 5]	[1 2 3 4 5]

3.2.7 problem statement

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details:** Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students:** Determine the total number of students in the dataset.
- **Print all student roll numbers:** Extract and print the roll numbers of all students.
- **Print Subject 1 marks:** Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2:** Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3:** Identify the highest marks in Subject 3.
- **Print all subject marks:** Display the marks of all students for each subject.

Find total marks of students: Compute the total marks for each student across all subjects..

- **Find the average marks of each student:** Compute the average marks for each student.
- **Find average marks of each subject:** Compute the average marks for all students in each subject.
- **Find average marks of Subject 1 and Subject 2:** Compute the average marks for Subject 1 and Subject 2.

- **Find average marks of Subject 1 and Subject 3:** Compute the average marks for Subject 1 and Subject 3.
- **Find the roll number of students who scored 24 marks in Subject 2:** Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- **Find the count of students who got less than 40 marks in Subject 1:** Count the number of students who scored less than 40 marks in Subject 1.
- **Find the count of students who got more than 90 marks in Subject 2:** Count the number of students who scored more than 90 marks in Subject 2.
- **Find the count of students who scored ≥ 90 in each subject:** Count the number of students who scored 90 or more marks in each subject.
- **Find the count of subjects in which each student scored ≥ 90 :** Determine how many subjects each student scored 90 or more marks in.
- **Print Subject 1 marks in ascending order:** Sort and print the marks of students in Subject 1 in ascending order.

- **Print students who scored between 50 and 90 in Subject 1:** Display students who scored marks between 50 and 90 in Subject 1.
- **Find index positions of students who scored 79 in subject 1:** Identify the index positions of students who scored exactly 79 marks in subject 1.

Code:

```
import numpy as np
```

```
a = np.loadtxt("Sample.csv", delimiter=',',  
skiprows=1)
```

```
# 1. Print all student details
```

```
print("All student Details:\n", a)
```

```
# 2. print total students
```

```
print("Total Students:", a.shape[0])
```

```
# 3. Print all student Roll numbers
```

```
print("All Student Roll Nos", a[:, 0])
```

```
# 4. Print subject 1 marks
```

```
print("Subject 1 Marks", a[:, 1])
```

```
# 5. print minimum marks of Subject 2
```



```
print("Min marks in Subject 2", np.min(a[:, 2]))  
  
# 6. print maximum marks of Subject 3  
print("Max marks in Subject 3", np.max(a[:, 3]))  
  
# 7. Print All subject marks  
print("All subject marks:", a[:, 1:])  
total = np.sum(a[:, 1:], axis=1)  
  
# 8. print Total marks of students  
print("Total Marks", total)  
  
#(no label in sample output)  
print(np.round(np.mean(a[:, 1:], axis=1), 1))  
  
# 9. print average marks of each student  
  
# 10. print average marks of each subject  
print("Average Marks of each subject",  
      np.round(np.mean(a[:, 1:], axis=0), 1))  
  
  
# 11. print average marks of S1 and S2  
print("Average Marks of S1 and S2",  
      np.round(np.mean(a[:, [1, 2]], axis=0), 1))  
  
# 12. print average marks of S1 and S3  
print("Average Marks of S1 and S3",  
      np.round(np.mean(a[:, [1, 3]], axis=0), 1) )
```

13. print Roll number who got maximum marks in Subject 3

```
print("Roll no who got maximum marks in Subject 3", a[np.argmax(a[:, 3]), 0])
```

14. print Roll number who got minimum marks in Subject 2

```
print("Roll no who got minimum marks in Subject 2", a[np.argmin(a[:, 2]), 0])
```

15. print Roll number who got 24 marks in Subject 2

```
print("Roll no who got 24 marks in Subject 2", a[a[:, 2] == 24, 0].reshape(-1,1))
```

16. print count of students who got marks in Subject 1 < 40

```
print("Count of students who got marks in Subject 1 < 40", np.sum(a[:, 1] < 40))
```

17. print count of students who got marks in Subject 2 > 90

```
print("Count of students who got marks in Subject 2 > 90:", np.sum(a[:, 2] > 90))
```

18. print count of students in each subject who got marks >= 90

```
print("Count of students in each subject who got  
marks >= 90:",
```

```
      np.sum(a[:, 1:] >= 90, axis=0))
```

```
# 19. print count of subjects in which each student  
got marks >= 90
```

```
print("Roll no:", a[:, 0])
```

```
print("Count of subjects in which student got  
marks >= 90:",
```

```
      np.sum(a[:, 1:] >= 90, axis=1))
```

```
# 20. Print S1 marks in ascending order
```

```
print(np.sort(a[:, 1]))
```

```
# 21. Print S1 marks >= 50 and <= 90
```

```
print(a[(a[:, 1] >= 50) & (a[:, 1] <= 90)])
```

```
print(a)
```

```
# 22. Print the index position of marks 79
```

```
print(np.where(a[:, 1] == 79))
```

Output:

Average time

0.014 s

14.00 ms



Maximum time

0.014 s

14.00 ms



✓ 1 out of 1 shown test case(s) passed

✓ Test case 1

14 ms



Expected output

Actual output

All · student · Details:
↕

All · student · Detai
ls: ↕

· [[301. · · 67. · · 77. · · 8
8.] ↕

· [[301. · · 67. · · 77.
· · 88.] ↕

· [302. · · 78. · · 88. · · 7
7.] ↕

· [302. · · 78. · · 88. ·
· 77.] ↕

· [303. · · 45. · · 56. · · 8
9.] ↕

· [303. · · 45. · · 56. ·
· 89.] ↕

· [304. · · 88. · · 98. · · 4
5.] ↕

· [304. · · 88. · · 98. ·
· 45.] ↕

·[305.···78.···88.···99.] [↗]	·[305.···78.···88.···99.] [↗]
·[306.···68.···78.···36.] [↗]	·[306.···68.···78.···36.] [↗]
·[307.···79.···12.···45.] [↗]	·[307.···79.···12.···45.] [↗]
·[308.···46.···45.···77.] [↗]	·[308.···46.···45.···77.] [↗]
·[309.···89.···32.···88.] [↗]	·[309.···89.···32.···88.] [↗]
·[310.···79.···24.···33.]] [↗]	·[310.···79.···24.···33.]] [↗]
Total·Students:·10 [↗]	Total·Students:·10 [↗]
All·Student·Roll·Nos·[301.·302.·303.·304.·305.·306.·307.·308.·309.·310.] [↗]	All·Student·Roll·Nos·[301.·302.·303.·304.·305.·306.·307.·308.·309.·310.] [↗]
Subject·1·Marks·[67.·78.·45.·88.·78.·68.·79.·46.·89.·79.] [↗]	Subject·1·Marks·[67.·78.·45.·88.·78.·68.·79.·46.·89.·79.] [↗]
Min·marks·in·Subject·2·12.0 [↗]	Min·marks·in·Subject·2·12.0 [↗]
Max·marks·in·Subject·3·99.0 [↗]	Max·marks·in·Subject·3·99.0 [↗]
All·subject·marks:·[[67.·77.·88.] [↗]	All·subject·marks:·[[67.·77.·88.] [↗]

· [78. · 88. · 77.] [↗]	· [78. · 88. · 77.] [↗]
· [45. · 56. · 89.] [↗]	· [45. · 56. · 89.] [↗]
· [88. · 98. · 45.] [↗]	· [88. · 98. · 45.] [↗]
· [78. · 88. · 99.] [↗]	· [78. · 88. · 99.] [↗]
· [68. · 78. · 36.] [↗]	· [68. · 78. · 36.] [↗]
· [79. · 12. · 45.] [↗]	· [79. · 12. · 45.] [↗]
· [46. · 45. · 77.] [↗]	· [46. · 45. · 77.] [↗]
· [89. · 32. · 88.] [↗]	· [89. · 32. · 88.] [↗]
· [79. · 24. · 33.] [↗]	· [79. · 24. · 33.] [↗]
Total Marks · [232. · 243. · 190. · 231. · 265. · 182. · 136. · 168. · 209. · 136.] [↗]	Total Marks · [232. · 243. · 190. · 231. · 265. · 182. · 136. · 168. · 209. · 136.] [↗]
[77.3 · 81. · · 63.3 · 77. · · 88.3 · 60.7 · 45.3 · 56. · · 69.7 · 45.3] [↗]	[77.3 · 81. · · 63.3 · 77. · · 88.3 · 60.7 · 45.3 · 56. · · 69.7 · 45.3] [↗]
Average Marks · of · each · subject · [71.7 · 59.8 · 67.7] [↗]	Average Marks · of · each · subject · [71.7 · 59.8 · 67.7] [↗]
Average Marks · of · S1 · and · S2 · [71.7 · 59.8] [↗]	Average Marks · of · S1 · and · S2 · [71.7 · 59.8] [↗]
Average Marks · of · S1 · and · S3 · [71.7 · 67.7] [↗]	Average Marks · of · S1 · and · S3 · [71.7 · 67.7] [↗]
Roll · no · who · got · maximum · marks · in · Subject · 3 · 305.0 [↗]	Roll · no · who · got · maximum · marks · in · Subject · 3 · 305.0 [↗]

Roll·no·who·got·mini- mum·marks·in·Subject ·2·307.0	Roll·no·who·got·m- inimum·marks·in·S ubject·2·307.0
Roll·no·who·got·24·m- arks·in·Subject·2· [[310.]]	Roll·no·who·got·2 4·marks·in·Subjec t·2·[[310.]]
Count·of·students·wh- o·got·marks·in·Subje ct·1·<·40·0	Count·of·students ·who·got·marks·in ·Subject·1·<·40·0
Count·of·students·wh- o·got·marks·in·Subje ct·2·>·90:·1	Count·of·students ·who·got·marks·in ·Subject·2·>·90:· 1
Count·of·students·in ·each·subject·who·go t·marks·>=·90:·[0·1· 1]	Count·of·students ·in·each·subject· who·got·marks·>=· 90:·[0·1·1]
Roll·no:·[301.·302.· 303.·304.·305.·306.· 307.·308.·309.·310.]	Roll·no:·[301.·30 2.·303.·304.·305. ·306.·307.·308.·3 09.·310.]
Count·of·subjects·in ·which·student·got·m arks·>=·90:·[0·0·0·1 ·1·0·0·0·0·0]	Count·of·subjects ·in·which·student ·got·marks·>=·90: ·[0·0·0·1·1·0·0·0 ·0·0]
[45.·46.·67.·68.·78. ·78.·79.·79.·88.·8 9.]	[45.·46.·67.·68.· 78.·78.·79.·79.·8 8.·89.]
[[301.··67.··77.··8 8.]	[[301.··67.··77.·· 88.]
·[302.··78.··88.··7	·[302.··78.··88.·

9.]	99.]
[306...68...78...36.]	[306...68...78...36.]
[307...79...12...45.]	[307...79...12...45.]
[309...89...32...88.]	[309...89...32...88.]
[310...79...24...33.]	[310...79...24...33.]
[[301...67...77...88.]	[[301...67...77...88.]
[302...78...88...77.]	[302...78...88...77.]
[303...45...56...89.]	[303...45...56...89.]
[304...88...98...45.]	[304...88...98...45.]
[305...78...88...99.]	[305...78...88...99.]
[306...68...78...36.]	[306...68...78...36.]
[307...79...12...45.]	[307...79...12...45.]
[308...46...45...77.]	[308...46...45...77.]
[309...89...32...88.]	[309...89...32...88.]
[310...79...24...33.]	[310...79...24...33.]
(array([6, 9], dtype=int32),)	(array([6, 9], dtype=int32),)

